BLOCKCHAİN-BASED ENROLLMENT MANAGEMENT SYSTEM FOR COLEGİO DE MARİNDUQUE

PRESENTED BY: JUNE RHEY MERCENE
MYLEN N. DAGSİL
DAPHNİ S. OLİVAR

MASTER İN İNFORMATİON TECHNOLOGY

BACKGROUND OF THE STUDY

- Colegio De Marinduque its enrollment process remains largely manual and fragmented.
- Students are required to fill out printed forms, submit physical documents such as report cards, PSA birth certificates, and identification photos, and undergo multiple steps, including aptitude tests and interviews.
- They must then proceed through various offices such as the registrar's office, the guidance office, and the cashier's office before enrollment is finalized.
- This multi-step process often results in long queues, repetitive verification, processing delays, and difficulty tracking student records and transactions, particularly during peak enrollment periods.

STATEMENT OF THE PROBLEM

- The enrollment process at Colegio De Marinduque (formerly Marinduque Technical & Vocational Institute) is inefficient for students and administrative staff due to its manual, fragmented system that requires multiple physical submissions, office visits, and payment steps, leading to delays and redundant verification.
- The lack of an integrated digital platform limits real-time validation and efficient record tracking, while the absence of blockchain or similar security mechanisms exposes records to tampering, loss, and unauthorized modification, with minimal traceability.

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RESEARCH QUESTİONS

Specifically, this study seeks to answer the following questions:

1. How can a blockchain-integrated enrollment management system be developed to digitize and centralize student registration, subject enrollment, payment verification, and academic record management while addressing inefficiencies in manual and fragmented processes?
2. How does blockchain technology improve the security, transparency, and immutability of enrollment records and transactions compared to traditional systems?
3. How effective and acceptable is the proposed system to end-users in terms of efficiency, accessibility, accuracy, usability, security, and overall performance?

OBJECTİVES OF THE STUDY

- **General Objective:** The project aims to develop a Blockchain-Integrated Enrollment Management System for Colegio de Marinduque to streamline student registration, subject enrollment, payment verification, and academic record management.

- **Specific Objectives:**

1. To design and develop a blockchain-integrated web-based enrollment management system that centralizes student registration, subject enrollment, payment processing, and academic record management while automating manual processes.
2. To implement blockchain technology to ensure secure, transparent, and immutable handling of enrollment records, payment transactions, and student data.
3. To evaluate the system's effectiveness and acceptability using ISO/IEC 25010 standards and user acceptance testing in terms of functionality, usability, efficiency, security, and overall performance.

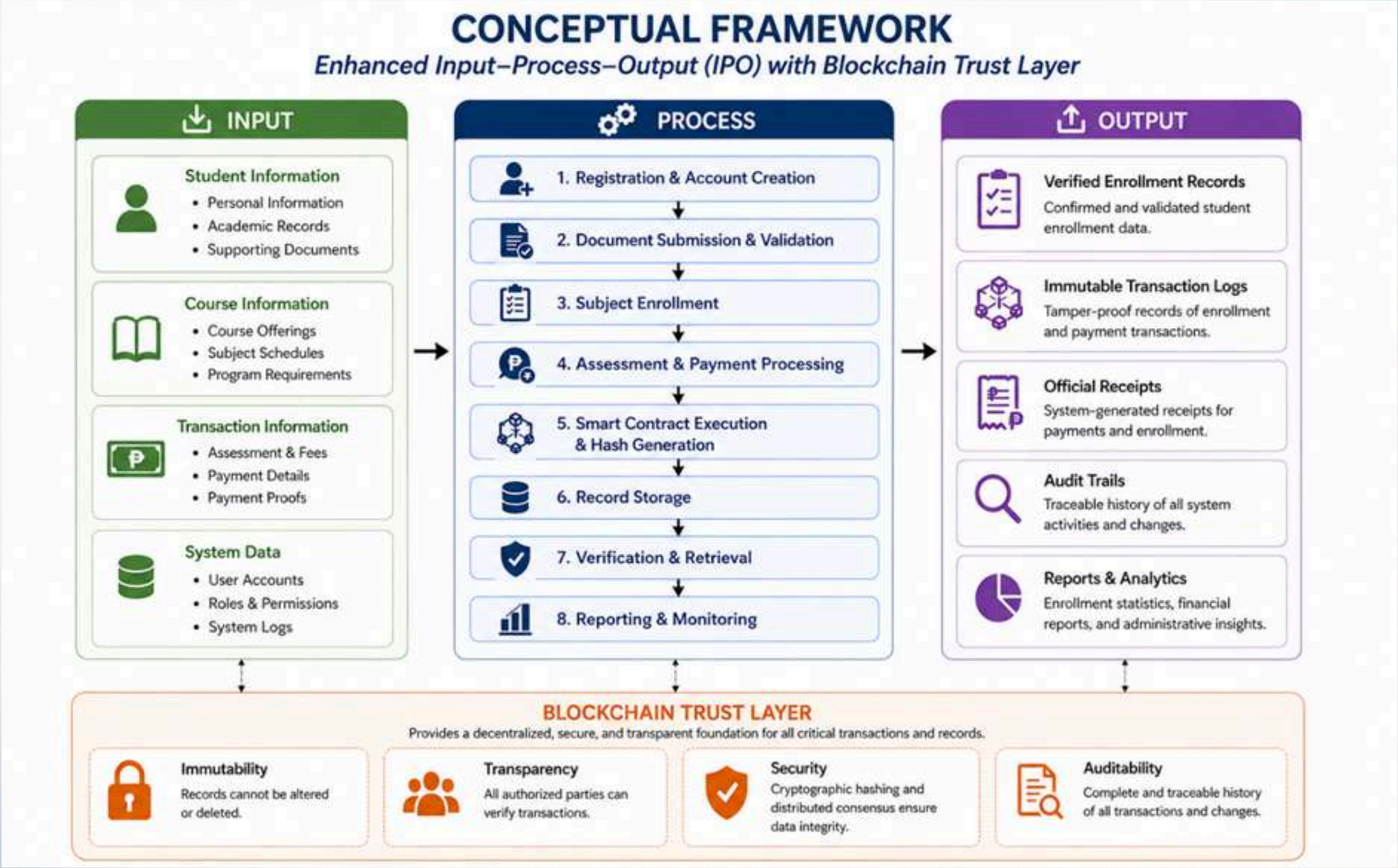
SİGNİFİCANCE OF THE STUDY

This study proposes a blockchain-integrated web-based enrollment management system designed to improve the efficiency, security, and reliability of enrollment processes at Colegio De Marinduque. The system aims to address the limitations of the current manual, fragmented procedures by providing a centralized, automated platform for enrollment operations. The following stakeholders are expected to benefit from the study:

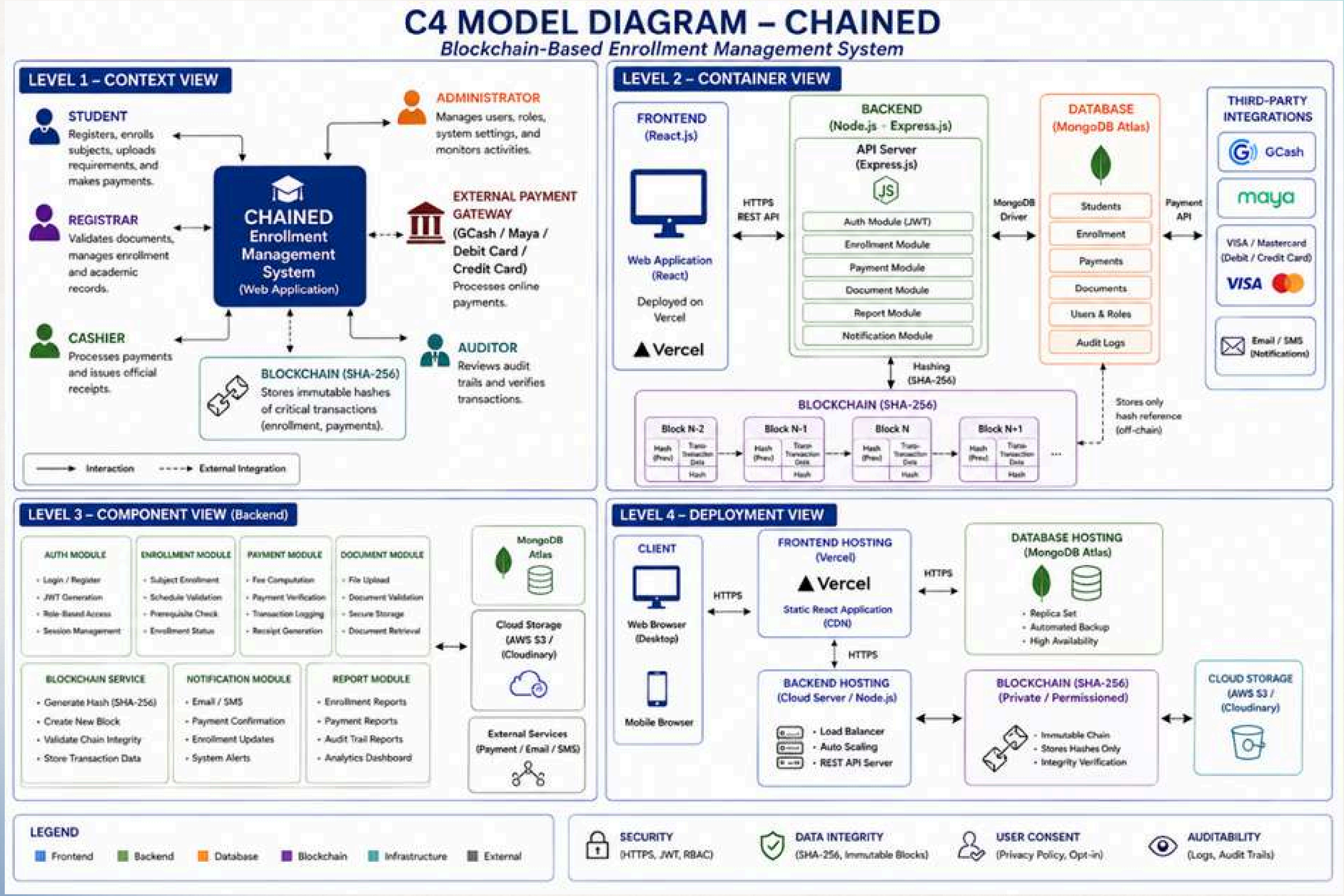
- Advisers
- Students
- Cashier
- Registrar

Overall, the study contributes to modernizing the enrollment system at Colegio De Marinduque by improving operational efficiency, strengthening data security, and enhancing service delivery for all stakeholders.

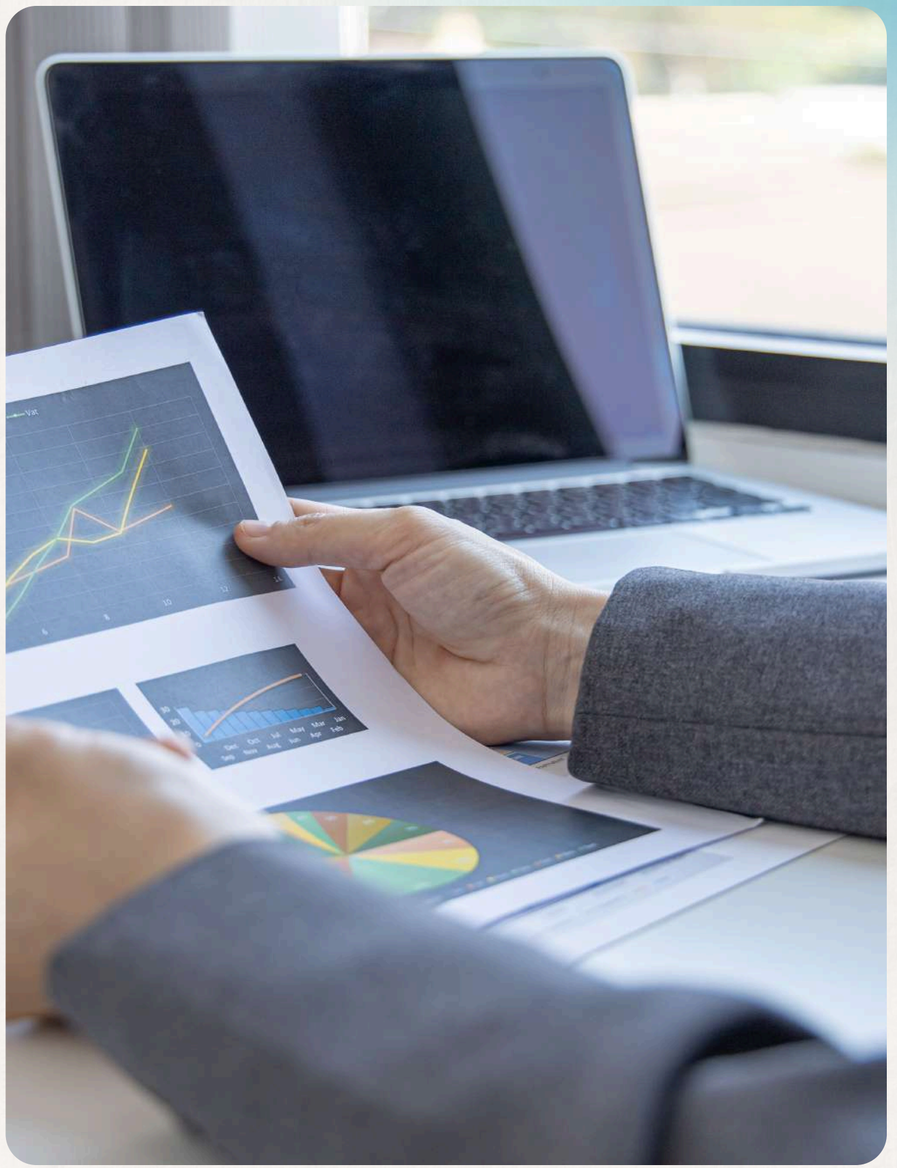
ARCHİTECTURAL / THEORETİCAL FRAMEWORK



PROPOSED SYSTEM ARCHİTECTURE



PRESENTATİON OF THE ENROLMENT SYSTEM



CONCLUSİON

The architectural intervention implemented in this study successfully addresses the limitations of traditional enrollment systems by improving processing efficiency, system scalability, and data integrity. While blockchain introduces some latency due to consensus and cryptographic operations, the trade-off is justified by the significant improvement in security and trustworthiness of records. The microservices approach further enhances system modularity and performance, making the system adaptable to varying enrollment demands.

THANK YOU
VERY MUCH